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Experiment No:	1
Subject:	Database Management Laboratory
Title:	Study of Open-Source Relational Databases: MySQL
Roll No.:	
Class:	TE (E&TC)
Division:	
Signature:	

PROGRESSIVE EDUCATION SOCIETY
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Group A- Database Programming Languages – SQL

Experiment No. -1

Title: Study of Open-Source Relational Databases: MySQL

Objective: To learn and understand open-source relational databases.

Hardware requirements: Any CPU with Pentium Processor or similar, 256 MB RAM or more, 1 GB Hard Disk or more.

Software requirements: Windows Operating System, MySQL

Theory:

A database-management system (DBMS) is a collection of interrelated data and a set of programs to access those data. The collection of data, usually referred to as the database, contains information relevant to an enterprise. The primary goal of a DBMS is to provide a way to store and retrieve database information that is both convenient and efficient. A relational database management system (RDBMS) is a program that lets you create, update, and administer a relational database. Most commercial RDBMS's use the Structured Query Language (SQL) to access the database, although SQL was invented after the development of the relational model and is not necessary for its use.

MySQL open-source RDBMS overview: MySQL is a popular open-source relational database management system (RDBMS) choice for web-based applications. Developers, database administrators use MySQL to build and manage next-generation web- and cloud-based applications. With most open-source RDBMS options, MySQL is available in several different editions and runs on Windows, OS X, Solaris, FreeBSD and other variants of Linux and Unix:

a) **MySQL Classic Edition**, available to only independent software vendors, OEMs and value-added resellers, is designed to be an embeddable database for read-intensive applications.
b) **MySQL Community Edition** is the free downloadable version of MySQL available under the GNU General Public License (GPL).
c) **MySQL Standard Edition** is the entry-level RDBMS offering for online transaction processing
d) **MySQL Enterprise Edition** adds advanced features, management tools (including OEM for MySQL) and technical support.
e) **MySQL Cluster Carrier Grade Edition** is designed for Web and cloud development.

Data types supported by MySQL open-source RDBMS:

MySQL data types include :

1. numeric types,
2. date and time types,
3. string types (including binary, character and Binary Large Object), and
4. spatial types.

Additionally, MySQL will map certain data types from other DBMSes to MySQL data types for easier portability.

Install MySQL on windows:

1. Extract the main archive to the desired install directory
2. Create an option file
3. Choose a MySQL server type
4. Initialize MySQL
5. Start the MySQL server
6. Secure the default user accounts

Install MySQL on Ubuntu 14.04: MySQL is an open-source database management system, commonly installed as part of the popular LAMP (Linux, Apache, MySQL, PHP/Python/Perl) stack. It uses a relational database and SQL (Structured Query Language) to manage its data. The short version of the installation is simple: update your package index, install the mysql-server package, and then run the included security and database initialization scripts.

Step 1 — Installing MySQL: There are two ways to install MySQL. You can either use one of the versions included in the APT package repository by default (which are 5.5 and 5.6), or you can install the latest version (currently 5.7) by manually adding MySQL's repository first. You can just use the mysql-server APT package, which just installs the latest version for your Linux distribution. To install MySQL this way, update the package index on your server and install the package with apt-get.

• sudo apt-get update

• sudo apt-get install mysql-server (You'll be prompted to create a root password during the installation. Choose a secure one and make sure you remember it, because you'll need it later.) Move on to step two from here.

Installing MySQL 5.5 or 5.6

If you want to install MySQL 5.5 or 5.6 specifically, the process is still very straight forward. First, update the package index on your server.

• sudo apt-get update Then, to install MySQL 5.5, (install the mysql-server-5.5 package).

• sudo apt-get install mysql-server-5.5 (you'll be prompted to create a root password during the installation. Choose a secure one and make sure you remember it, because you'll need it later.)

Step 2 — Configuring MySQL: First, you'll want to run the included security script. This changes some of the less secure default options for things like remote root logins and sample users.

• sudo mysql_secure_installation

This will prompt you for the root password you created in step one. You can press ENTER to accept the defaults for all the subsequent questions, with the exception of the one that asks if you'd like to change the root password. You just set it in step one, so you don't have to change it now. Next, Initialize the MySQL data directory, which is where MySQL stores its data. How you do this depends on which version of MySQL you're running. You can check your version of MySQL with the following command.

• mysql –version If you're using a version of MySQL earlier than 5.7.6, you should initialize the data directory by running mysql_install_db.

• sudo mysql_install_db

Step 3 — Testing MySQL: Regardless of how you installed it, MySQL should have started running automatically. To test this, check its status.

• service mysql status

If MySQL isn't running, you can start it with

• sudo service mysql start.

For an additional check, you can try connecting to the database using the mysqladmin tool, which is a client that lets you run administrative commands. For JES'ITMR DBMS LABORATORY example, this command says to connect to MySQL as root (-u root), prompt for a password (-p), and return the version.

• mysqladmin -p -u root version

Step 4 — Log in MySQL: Use following command to log in to MySQL in terminal, write

• sudo mysql –u root –p You will be prompted for password.

After log in you can see mysql> prompt to execute queries

RDBMS Terminology:

Before we proceed to explain database system, let's revise few definitions related to database.

Database: A database is a collection of tables, with related data.

Table: A table is a matrix with data. A table in a database looks like a simple spreadsheet.

Column: One column (data element) contains data of one and the same kind, for example the Column postcode.

Row: A row (tuple, entry or record) is a group of related data, for example the data of one subscription.

Redundancy: Storing data twice, redundantly to make the system faster.

Primary Key: A primary key is unique. A key value cannot occur twice in one table. With a key, you can find at most one row.

Foreign Key: A foreign key is the linking pin between two tables.

Compound Key: A compound key (composite key) is a key that consists of multiple columns, because one column is not sufficiently unique.

Index: An index in a database resembles an index at the back of a book.

Referential Integrity: Referential Integrity makes sure that a foreign key value always points to an existing row. is a fast, easy-to-use RDBMS being used for many small and big businesses. is developed, marketed, and supported by My SQLAB, which is a Swedish company. is becoming so popular because of many good reasons:

• Released under an open-source license. So, you have nothing to pay to use it.

• Very powerful program. It handles a large subset of the functionality of the most expensive and powerful database packages.

• Uses a standard form of the well-known SQL data language.

•Works on many operating systems and with many languages including PHP, PERL, C, C++, JAVA, etc.

• Works very quickly and works well even with large datasets.

• Friendly to PHP, the most appreciated language for web development.

• Supports large databases, upto 50 million rows or more in a table. The default file size limit for a table is 4GB, but you can increase this (if your operating system can handle it) to a theoretical limit of 8 million tera bytes (TB).

• Customizable. The open-source GPL license allows programmers to modify the software to fit their own specific environments.

SQL General Data Types

Each column in a database table is required to have a name and a data type.

SQL developers have to decide what types of data will be stored inside each table column when creating a SQL table. The data type is a label and a guideline for SQL to understand what type of data is expected inside of each column, and it also identifies how SQL will interact with the stored data.

The following table lists the general data types in SQL:

Data Type	Description
String Datatype	
CHAR (size)	A FIXED length string (can contain letters, numbers, and special characters). The *size* parameter specifies the column length in characters - can be from 0 to 255. Default is 1
VARCHAR (size)	A VARIABLE length string (can contain letters, numbers, and special characters). The *size* parameter specifies the maximum column length in characters - can be from 0 to 65535
BINARY (size)	Equal to CHAR() but stores binary byte strings. The *size* parameter specifies the column length in bytes. Default is 1
VARBINARY (size)	Equal to VARCHAR(), but stores binary byte strings. The *size* parameter specifies the maximum column length in bytes.
TINYBLOB	For BLOBs (Binary Large OBjects). Max length: 255 bytes
TINYTEXT	Holds a string with a maximum length of 255 characters
TEXT (size)	Holds a string with a maximum length of 65,535 bytes
BLOB (size)	For BLOBs (Binary Large OBjects). Holds up to 65,535 bytes of data
MEDIUMTEXT	Holds a string with a maximum length of 16,777,215 characters
MEDIUMBLOB	For BLOBs (Binary Large OBjects). Holds up to 16,777,215 bytes of data
LONGTEXT	Holds a string with a maximum length of 4,294,967,295 characters
LONGBLOB	For BLOBs (Binary Large OBjects). Holds up to 4,294,967,295 bytes of data
ENUM (val1, val2, val3, ...)	A string object that can have only one value, chosen from a list of possible values. You can list up to 65535 values in an ENUM list. If a value is inserted that is not in the list, a blank value will be inserted. The values are sorted in the order you enter them
Numeric Data Type	
BIT (size)	A bit-value type. The number of bits per value is specified in size. The size parameter can hold a value from 1 to 64. The default value for size is 1.

TINYINT (size)	A very small integer. Signed range is from -128 to 127. Unsigned range is from 0 to 255. The size parameter specifies the maximum display width (which is 255)
BOOL	Zero is considered as false, nonzero values are considered as true.
BOOLEAN	Equal to BOOL
SMALLINT (size)	A small integer. Signed range is from -32768 to 32767. Unsigned range is from 0 to 65535. The size parameter specifies the maximum display width (which is 255)
MEDIUMINT (size)	A medium integer. Signed range is from -8388608 to 8388607. Unsigned range is from 0 to 16777215. The size parameter specifies the maximum display width (which is 255)
INT (size)	A medium integer. Signed range is from -2147483648 to 2147483647. Unsigned range is from 0 to 4294967295. The size parameter specifies the maximum display width (which is 255)
INTEGER (size)	Equal to INT (size)
BIGINT (size)	A large integer. Signed range is from -9223372036854775808 to 9223372036854775807. Unsigned range is from 0 to 18446744073709551615. The size parameter specifies the maximum display width (which is 255)
FLOAT (size, d)	A floating point number. The total number of digits is specified in size. The number of digits after the decimal point is specified in the d parameter. This syntax is deprecated in MySQL 8.0.17, and it will be removed in future MySQL versions
FLOAT(p)	A floating point number. MySQL uses the p value to determine whether to use FLOAT or DOUBLE for the resulting data type. If p is from 0 to 24, the data type becomes FLOAT (). If p is from 25 to 53, the data type becomes DOUBLE ()
DOUBLE (size, d)	A normal-size floating point number. The total number of digits is specified in size. The number of digits after the decimal point is specified in the d parameter
DOUBLE PRECISION (size, d)	

DECIMAL (size, d)	An exact fixed-point number. The total number of digits is specified in size. The number of digits after the decimal point is specified in the d parameter. The maximum number for size is 65. The maximum number for d is 30. The default value for size is 10. The default value for d is 0.
DEC (size, d)	Equal to DECIMAL (size,d)
Date and Time Data Types	
DATE	A date. Format: YYYY-MM-DD. The supported range is from '1000-01-01' to '9999-12-31'
DATETIME (fsp)	A date and time combination. Format: YYYY-MM-DD hh:mm:ss. The supported range is from '1000-01-01 00:00:00' to '9999-12-31 23:59:59'. Adding DEFAULT and ON UPDATE in the column definition to get automatic initialization and updating to the current date and time
TIMESTAMP (fsp)	A timestamp. TIMESTAMP values are stored as the number of seconds since the Unix epoch ('1970-01-01 00:00:00' UTC). Format: YYYY-MM-DD hh:mm:ss. The supported range is from '1970-01-01 00:00:01' UTC to '2038-01-09 03:14:07' UTC. Automatic initialization and updating to the current date and time can be specified using DEFAULT CURRENT_TIMESTAMP and ON UPDATE CURRENT_TIMESTAMP in the column definition
TIME (fsp)	A time. Format: hh:mm:ss. The supported range is from '-838:59:59' to '838:59:59'
YEAR	A year in four-digit format. Values allowed in four-digit format: 1901 to 2155, and 0000. MySQL 8.0 does not support year in two-digit format.

Conclusion: